

5 MARKET OVERVIEW

5.1 INTRODUCTION

The microbiome sequencing services market is estimated to grow at a CAGR of 14.3% during the forecast period. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

5.2 MARKET DYNAMICS

FIGURE 18 MICROBIOME SEQUENCING SERVICES MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

 DRIVERS	▪ Increasing demand for metagenomic sequencing ▪ Growing focus on human microbiome therapeutics development ▪ High cost of advanced infrastructure for sequencing
 RESTRAINTS	▪ Dearth of skilled personnel
 OPPORTUNITIES	▪ Expanding applications of microbiome sequencing in therapeutics development
 CHALLENGES	▪ Issues related to sample preparation/isolation of microbes for sequencing

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 4 MICROBIOME SEQUENCING SERVICES MARKET: IMPACT ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

FACTOR	IMPACT		
	2-3 YEARS	3-5 YEARS	5-10 YEARS
DRIVERS			
Increasing demand for metagenomic sequencing	High	High	High
Growing focus on human microbiome therapeutics development	Medium	High	Medium
High cost of advanced infrastructure for sequencing	Medium	Medium	Medium
RESTRAINTS			
Dearth of skilled personnel	High	Medium	Medium
OPPORTUNITIES			
Expanding applications of microbiome sequencing in therapeutics development	Medium	Medium	High
CHALLENGES			
Issues related to sample preparation/isolation of microbes for sequencing	Medium	Low	Low

5.2.1 DRIVERS

5.2.1.1 Increasing demand for metagenomic sequencing

Metagenomic sequencing is a target-independent approach that offers a comprehensive view of pathogenic agents and enables the detection of common and unexpected pathogens in samples. In recent years, there has been a significant increase in the demand for metagenomic sequencing in the field of microbiome research owing to its wide array of applications, such as disease diagnosis, drug discovery, and precision medicine. Recent developments in metagenomic sequencing technologies, such as improvements in sequencing accuracy, scalability, and data analysis, support the use of metagenomic sequencing. The development of long-read sequencing, single-molecule real-time sequencing, and nanopore sequencing has been particularly helpful in the metagenomic sequencing market. The demand for long-read sequencing-based metagenomics has increased in samples that are less complex, involve host-associated microbiomes, and comprise fewer microbial strains than microbiomes in marine environments, sediment, and soil. Such technologies are expected to drive the demand for microbiome sequencing services.

Over the years, the demand for precision medicine has increased. Precision medicine involves tailoring medical treatments and interventions to an individual's unique characteristics, including their microbiome. Metagenomic sequencing services are helpful in gathering essential information for personalized medicine, as well as identifying specific microbial signatures associated with disease risk and specific treatment responses. Metagenomic sequencing is also being used to study the role of microbiomes in cancer development, progression, and response to treatment. As per a study published in the *Lancet* in 2022, the gut microbiome of patients with metastatic melanoma can influence their response to immune checkpoint inhibitors. The study identified specific microbial species that were associated with better response rates.